Algorithmen für KI und Data Science II
SoSe 2025

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Exercise Sheet 06

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Total points: 10

Please upload your solutions to WueCampus as a scanned document (image format or pdf), a typesetted PDF document, and/or as a `jupyter` notebook.

1. **Bayesian probability**

   (a) Given the scenario where a doctor is attending to a sick child, the doctor possesses prior knowledge that 90% of sick children in that neighborhood have the flu, while the remaining 10% are affected by measles. Let F represent the event of a child being sick with the flu, and M represent the event of a child being sick with measles. It is assumed, for simplicity, that the set of all possible outcomes (denoted as Ω) comprises only the presence of flu or measles; no other illnesses exist in that neighborhood. 2P

   One notable symptom of measles is the presence of a rash (denoted as event R). Assuming that the probability of having a rash if a child has measles is $P(R|M) = 0.95$. However, in some cases, children with the flu may also develop a rash, with a probability of $P(R|F) = 0.08$.

   In this situation, upon examining the child, the doctor discovers the presence of a rash. The question arises: What is the probability that the child has measles?

   (b) Approximately 50% of all emails are estimated to be spam emails. To combat this issue, certain software has been implemented to filter out these spam emails and prevent them from reaching your inbox. A specific software brand boasts a remarkable 99% accuracy in detecting spam emails, while maintaining a low false positive rate of 5% (non-spam emails incorrectly identified as spam). 2P

   Now, if an email is flagged as spam, what is the probability that it is actually a non-spam email?

2. **Statistical inference**

   (a) Utilizing the Linear Regression model from the Sklearn package in Python, carry out the following experiment: 2P

   a) Import the required data using the following commands:

   ```
   from sklearn.datasets import load_diabetes
   data = load_diabetes()
   ```

   b) Divide the dataset into training and test sets using the following procedure:

   ```
   from sklearn.model_selection import train_test_split
   X_train, X_test, y_train, y_test = train_test_split(
           data['data'],
           data['target'],
           test_size=0.33,
           random_state=42
           )
   ```

   c) Apply the Linear Regression model and display the R^2 score, intercept, and slope.

d) Compute the covariance matrix.

(b) The Central Limit Theorem states that as the sample size increases, the distribution of sample means approaches a normal distribution. Consider a scenario where we generate 1000 random samples from a uniform distribution between 0 and 1. We then calculate the mean of each sample. What would be the expected shape of the plot showing the sample means against the sample size? 2P

(c) Consider a coin tossing scenario where a coin is tossed several times until the first head appears. The probability p of head is unknown. 2P

(i) Starting from a uniform prior distribution of p, calculate the posterior distribution of p after the coin is tossed n times until the first head.

(ii) Use maximum and mean a posteriori estimation to obtain an estimation of the model parameter $\hat{p}$

(iii) Plot the posterior distribution of p when the coin is tossed 5 times until the first head.